THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · HARD LEVEL CHAPTER TEST SET 1

Chapter 2: **Invisible Living World: Beyond Our Naked Eye** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Assertion (A): Animal cells burst more easily than plant cells in very dilute surroundings. Reason (R): Animal cells have a rigid cell wall. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

2. A bacterium of length 2 micrometres is viewed with total magnification of 1000x. Its image size is: 1

(a) 2000 mm **(b)** 2 mm **(c)** 0.2 mm **(d)** 20 mm

3. A cell loses its nucleus. The most probable result is that the cell: 1

(a) becomes a plant cell **(b)** stops functioning properly
(c) forms chloroplasts **(d)** grows faster

4. Assertion (A): Chloroplasts are found in plant cells but not in animal cells. Reason (R): Plants make their own food while animals depend on others. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

5. A student observes an onion peel with 10x eyepiece and 10x objective, and the cell length in the picture is 5 mm. The real length is: 1

(a) 0.005 mm **(b)** 5 mm **(c)** 0.05 mm **(d)** 0.5 mm

6. A dough is kept (i) in a warm place, (ii) in a fridge. Dough (i) rises more because: 1

(a) yeast dies in warmth **(b)** the cooler dough releases more CO2
(c) yeast is more active at warm temperature **(d)** the fridge makes yeast grow faster

7. Assertion (A): Antibiotics are not given for viral infections such as influenza. Reason (R): Viruses do not have the cell structures targeted by antibiotics. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

8. A cell has chloroplasts, but the question says the cell is from an underground part of a plant. The conclusion should be: 1

(a) check, since chloroplasts are rare underground
(b) it is an animal cell
(c) chloroplasts need no light
(d) all roots are green

9. Which is NOT a feature of a unicellular organism? 1

(a) cytoplasm **(b)** tissues

(c) cell membrane **(d)** nucleus (in protozoa)

10. Assertion (A): A red blood cell can carry more oxygen because it has no nucleus. Reason (R): Haemoglobin takes up the space gained. 1

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. A cell viewed at 500x magnification appears 5 mm long. Find the actual length in micrometres. 2

12. Why is Amoeba able to change its shape but a plant cell cannot? 2

13. Why is the cell called the 'building block of life'? 2

14. Why did Hooke see only walls in cork cells and not the contents? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. An experiment: a student places an onion peel in concentrated salt solution. Cell contents shrink from the wall. What does this show about the cell wall and membrane? 3

16. Name the processes carried out by Amoeba that are done by organs in a human. 3

17. A farmer sprays an antibiotic on a crop disease caused by a fungus and it fails. Why? 3

OR

Differentiate between a virus and a bacterium based on size, structure and treatment. 3

18. Explain what would happen to a multicellular organism if its cells did not specialise. 3

OR

Explain why a human egg cell, though a single cell, is visible to the naked eye while bacteria are not. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Describe an experiment to study the structure of a plant cell and an animal cell, and compare the observations in a table. 5

OR

Explain why the cell surface area to volume ratio matters. Why do cells divide instead of growing without limit? 5

20. A cell is magnified 800x using a 10x eyepiece. Find the objective power. If the image is 4 mm, find the actual size in micrometres. 5

OR

A student prepared a slide of curd and saw many tiny rod-like cells. Explain how they are formed, are they useful, and how to make curd at home. 5

END OF QUESTION PAPER — Total: 40 Marks